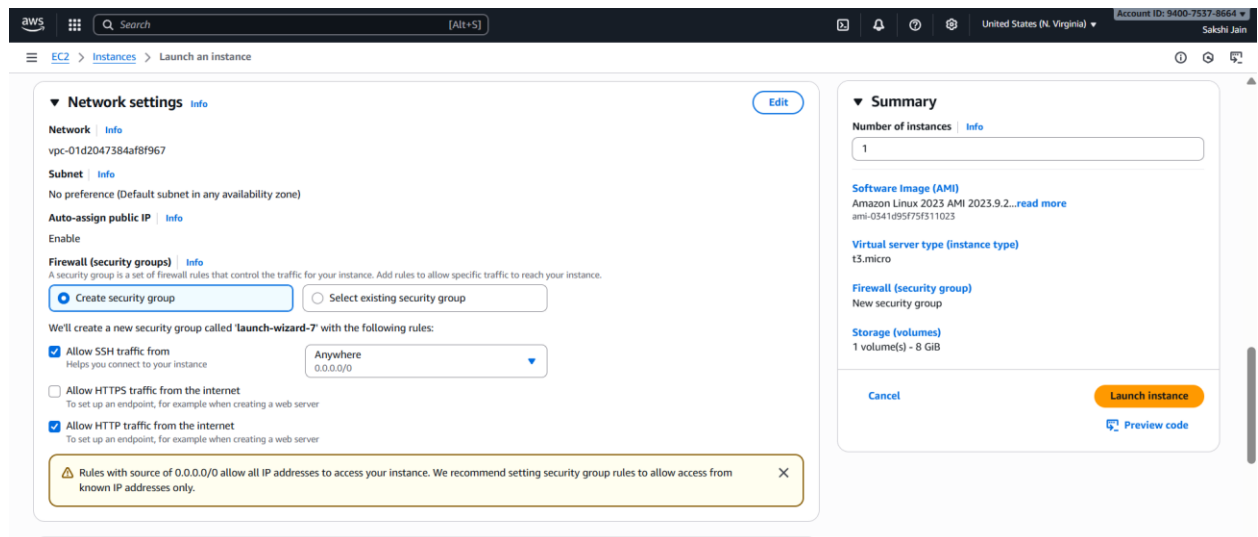


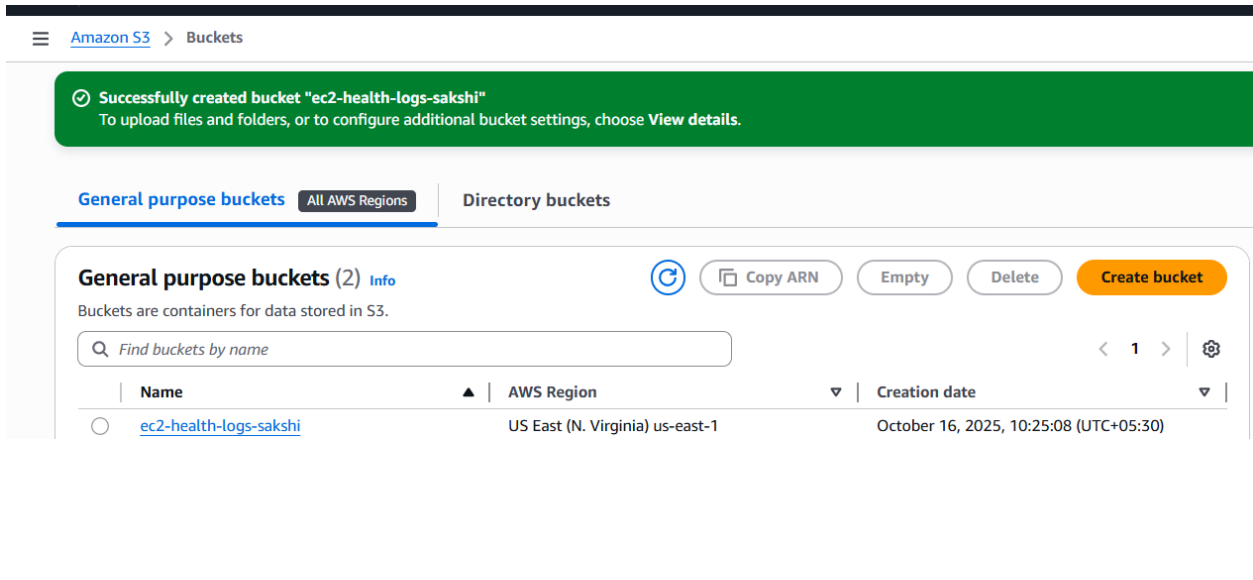
STEP 1: Create an EC2 Instance

1. Login to AWS Console → <https://aws.amazon.com/console>
2. Search for "EC2" → open it.
3. Click Launch Instance
 - Name: HealthCheckInstance
 - AMI: Amazon Linux 2
 - Instance type: t2.micro (Free Tier)
4. Key Pair → create a new one → download .pem file.
5. Network Settings:
 - Allow SSH (port 22)
 - Allow HTTP (port 80)
6. Click Launch Instance 



STEP 2: Create an S3 Bucket for Logs

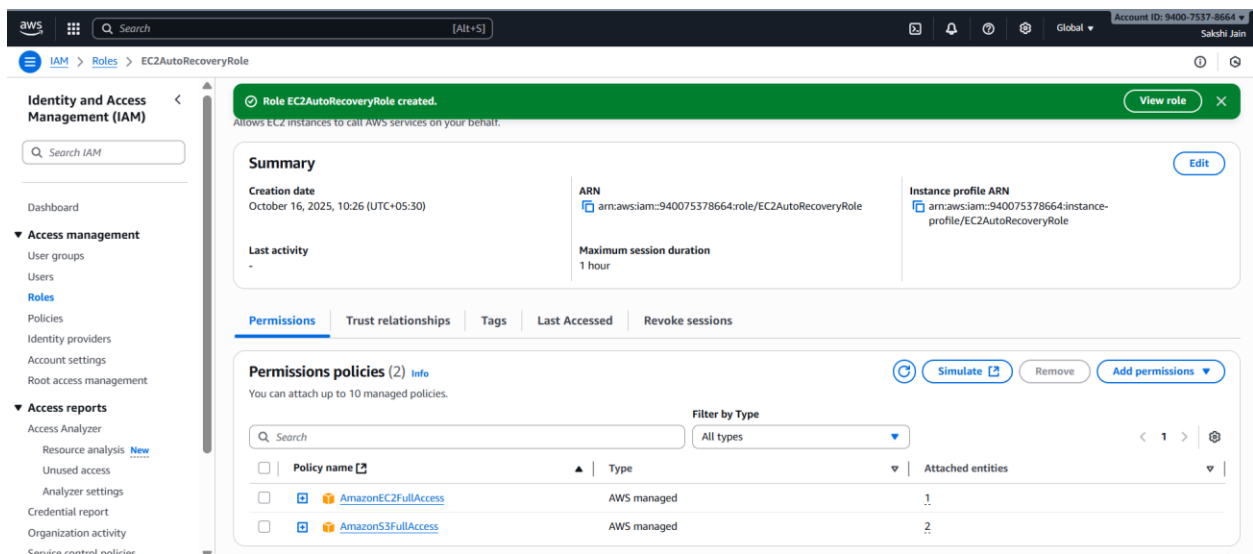
1. Go to S3 → Create bucket
 - Name: ec2-health-logs-sakshi
 - Region: same as your EC2 region
2. Keep all defaults → Create bucket

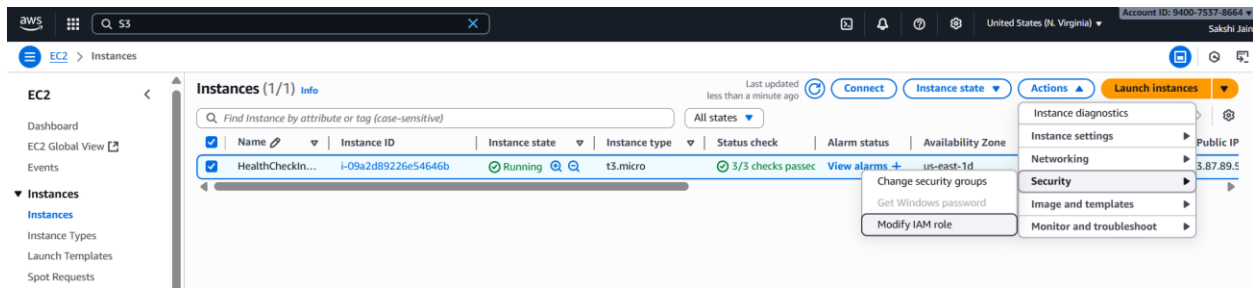


STEP 3: Create IAM Role for EC2

This gives EC2 permission to restart itself & write logs to S3.

1. Go to IAM → Roles → Create role
2. Choose AWS Service → EC2
3. Click Next and attach these policies:
 - AmazonS3FullAccess
 - AmazonEC2FullAccess
4. Name the role: EC2AutoRecoveryRole
5. Click Create role
6. Go to EC2 → Instances → Select your instance → Actions → Security → Modify IAM Role
 - Attach the role EC2AutoRecoveryRole 

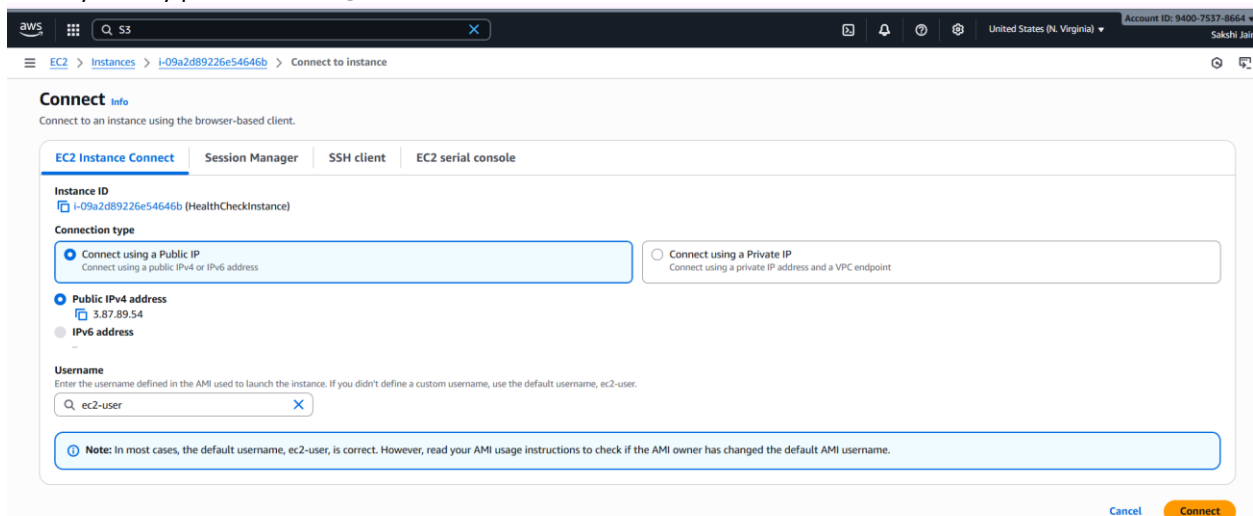




STEP 4: Connect to EC2 Instance

From your terminal / command prompt:

```
ssh -i "your-key.pem" ec2-user@<EC2-Public-IP>
```



STEP 5: Install AWS CLI (if not already)

```
sudo yum update -y
sudo yum install aws-cli -y
```

Check:

```
aws --version
```

STEP 6: Create the Health Check Script

Inside EC2 terminal:

vim healthcheck.sh

Paste this script 🖱️

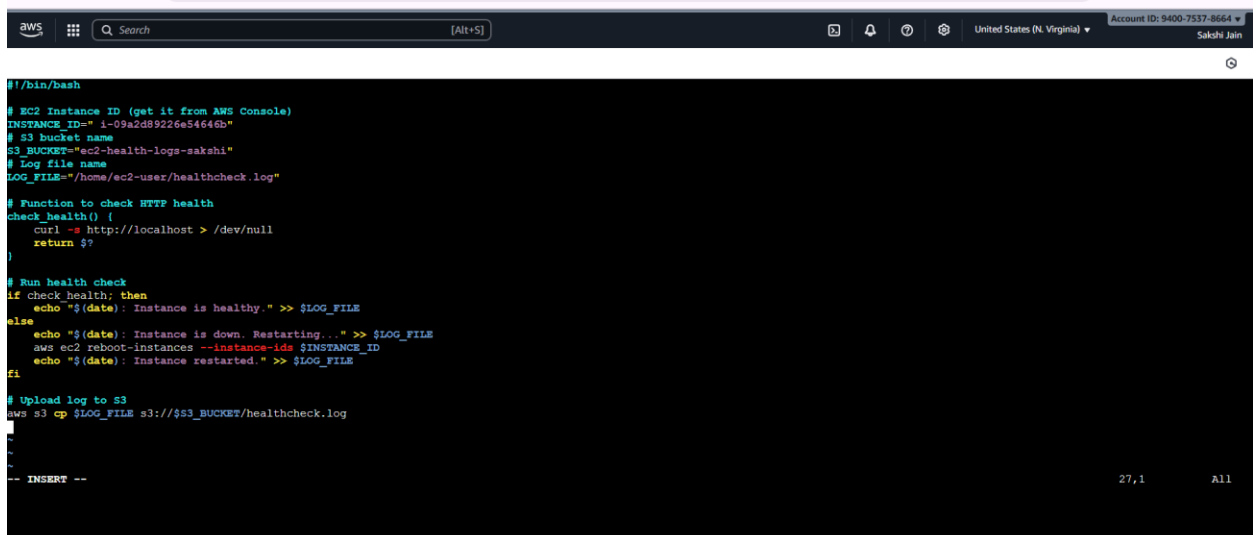
```
#!/bin/bash

# EC2 Instance ID (get it from AWS Console)
INSTANCE_ID="i-09a2d89226e54646b"
# S3 bucket name
S3_BUCKET="ec2-health-logs-sakshi"
# Log file name
LOG_FILE="/home/ec2-user/healthcheck.log"

# Function to check HTTP health
check_health() {
    curl -s http://localhost > /dev/null
    return $?
}

# Run health check
if check_health; then
    echo "$(date): Instance is healthy." >> $LOG_FILE
else
    echo "$(date): Instance is down. Restarting..." >> $LOG_FILE
    aws ec2 reboot-instances --instance-ids $INSTANCE_ID
    echo "$(date): Instance restarted." >> $LOG_FILE
fi

# Upload log to S3
aws s3 cp $LOG_FILE s3://$S3_BUCKET/healthcheck.log
```



```
aws
Q Search [Alt+S] United States (N. Virginia) Account ID: 9400-7537-8664 Sakshi Jain

#!/bin/bash

# EC2 Instance ID (get it from AWS Console)
INSTANCE_ID="i-09a2d89226e54646b"
# S3 bucket name
S3_BUCKET="ec2-health-logs-sakshi"
# Log file name
LOG_FILE="/home/ec2-user/healthcheck.log"

# Function to check HTTP health
check_health() {
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if check_health; then
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else
    echo "$(date): Instance is down. Restarting..." >> $LOG_FILE
    aws ec2 reboot-instances --instance-ids $INSTANCE_ID
    echo "$(date): Instance restarted." >> $LOG_FILE
fi

# Upload log to S3
aws s3 cp $LOG_FILE s3://$S3_BUCKET/healthcheck.log

-- INSERT --
```

STEP 7: Make the Script Executable

```
chmod +x healthcheck.sh
```

```

aws
[Alt+S]
United States (N. Virginia)
Account ID: 9400-7537-8664
Sakshi Jain

Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-27-11 ~]$ sudo -i
[root@ip-172-31-27-11 ~]# vim healthcheck.sh
[root@ip-172-31-27-11 ~]# chmod +x healthcheck.sh
[root@ip-172-31-27-11 ~]# yum install cronie -y
Amazon Linux 2023 Kernel Livepatch repository 187 kB/s | 26 kB 00:00
Dependencies resolved.

Package Architecture Version Repository Size
Installing:
cronie x86_64 1.5.7-1.amzn2023.0.2 amazonlinux 115 k
Installing dependencies:
cronie-anacron x86_64 1.5.7-1.amzn2023.0.2 amazonlinux 32 k

Transaction Summary
Install 2 Packages

Total download size: 147 k
Installed size: 341 k
Downloading Packages:
(1/2): cronie-anacron-1.5.7-1.amzn2023.0.2.x86_64.rpm 907 kB/s | 32 kB 00:00
(2/2): cronie-1.5.7-1.amzn2023.0.2.x86_64.rpm 2.7 MB/s | 115 kB 00:00
Total 1.7 MB/s | 147 kB 00:00
Running transaction check

```

STEP 8: Schedule the Script Automatically (Cron Job)

```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-27-11 ~]$ sudo -i
[root@ip-172-31-27-11 ~]# vim healthcheck.sh
[root@ip-172-31-27-11 ~]# chmod +x healthcheck.sh
[root@ip-172-31-27-11 ~]# yum install cronie -y
Amazon Linux 2023 Kernel Livepatch repository 187 kB/s | 26 kB 00:00
Dependencies resolved.

Package Architecture Version Repository Size
Installing:
cronie x86_64 1.5.7-1.amzn2023.0.2 amazonlinux 115 k
Installing dependencies:
cronie-anacron x86_64 1.5.7-1.amzn2023.0.2 amazonlinux 32 k

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Install 2 Packages

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(2/2): cronie-1.5.7-1.amzn2023.0.2.x86_64.rpm 2.7 MB/s | 115 kB 00:00
Total 1.7 MB/s | 147 kB 00:00
Running transaction check
```

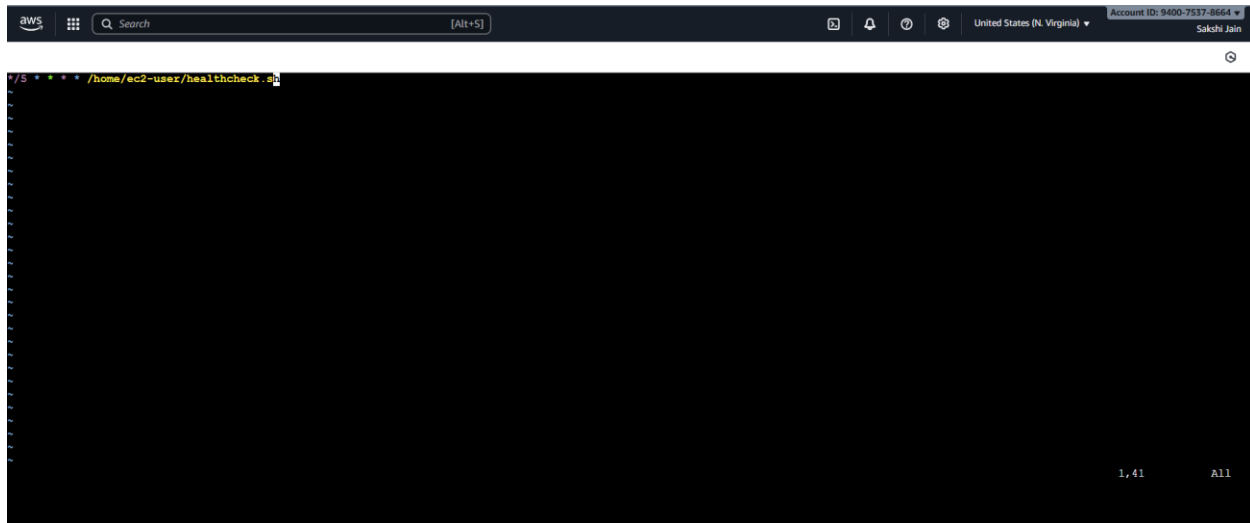
Open crontab:

```
crontab -e
```

Add this line (run every 5 minutes):

```
* /5 * * * * /home/ec2-user/healthcheck.sh
```

Save and exit.



STEP 9: Testing (How to Check Final Output)

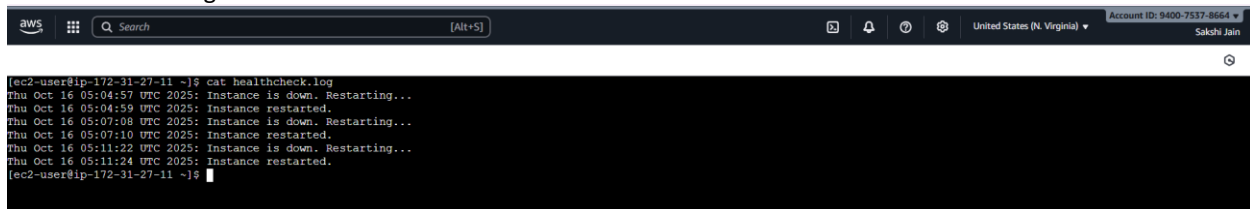
Test 1: Normal Condition

Run manually first: Run with Root user and Need to Start and Stop HTTPD service first.

```
./healthcheck.sh
```

Then check: from Ec2 user(local User)

```
cat healthcheck.log
```



You should see:

Thu Oct 16 10:00:00 UTC 2025: Instance is healthy.

Also check your S3 bucket — it will have healthcheck.log.

Test 2: Simulate “Instance Down”

To test recovery:

1. Stop your web service (simulate failure):
 2. `sudo systemctl stop httpd`
 3. Run script again:
 4. `./healthcheck.sh`
 5. It will detect “down”, restart EC2, and log this:
 6. Thu Oct 16 10:05:00 UTC 2025: Instance is down. Restarting...
 7. Thu Oct 16 10:05:15 UTC 2025: Instance restarted.
 8. Check S3 bucket again → you’ll see updated logs uploaded automatically.
-

FINAL OUTPUT (Summary)

EC2 instance automatically checked
Automatically restarted when down
Logs stored to S3
Continuous monitoring via cron